Knowledge-Based Systems 280 (2023) 111008

Contents lists available at ScienceDirect

Knowledge-Based Systems

journal homepage: www.elsevier.com/locate/knosys

An evolutionary filter approach to feature selection in classification for both single- and multi-objective scenarios

Emrah Hancer [a,b,*], Bing Xue [a], Mengjie Zhang [a]

[a] Center for Data Science and Artificial Intelligence & School of Engineering and Computer Science, Victoria University of Wellington, Wellington 6140, New Zealand

[b] Department of Software Engineering, Mehmet Akif Ersoy University, Burdur 15039, Turkey

ARTICLE INFO

Keywords:
Differential evolution
Neighborhood component analysis
Multi-objective optimization
Classification

ABSTRACT

The high-dimensional datasets in various domains, such as text categorization, information retrieval and bioinformatics, have highlighted the importance of feature selection in data mining. Despite the numerous existing approaches to feature selection, there is still a need for further research in this field. In this paper, we propose an evolutionary filter feature selection approach that can be used for both single- and multi-objective scenarios by introducing an objective function inspired by Neighborhood Component Analysis (NCA)-based method and then integrating it into the differential evolution framework. The proposed approach applicable to two scenarios aims to identify an optimal feature subset through an evolutionary search process that maximizes class separation while minimizing the dimensionality. Through comprehensive experimental studies conducted on diverse datasets, the results show that the proposed approach outperforms recently proposed evolutionary information-theoretic, rough set-based and state-of-the-art feature selection approaches in both scenarios. Notably, this study is the first to integrate an NCA-based strategy into an evolutionary feature selection approach. Furthermore, you can access the source code of this approach at https://github.com/ehancer06/DENCA this link.

1. Introduction

Classification is a fundamental task in machine learning, with significant implications across various domains. In a typical classification scenario, a classifier is trained using a labeled dataset, where each sample is associated with a specific class label. The training process aims to establish the connection between the variables, also known as features or properties, of the samples and their respective class labels. Once the classifier is trained, it can be employed to predict the class labels of new, unseen test samples. It is crucial to note that the effectiveness of the classifier is directly influenced by the quality and quantity of the features utilized in the training process.

High-dimensional datasets often contain redundant or irrelevant features that may include incorrect or misleading information about the class labels. The presence of such redundant, irrelevant and noisy features hampers the identification of relevant or discriminating features, leading to that a classifier learned from these features becomes less efficient and accurate [1]. Furthermore, the curse of dimensionality exacerbates the training process of classifiers when the number of features increases exponentially. This necessitates a significantly larger number of samples for effective training [2]. In the context of high-dimensional data, feature selection plays a critical role. It enables the reduction of feature dimensions and enhances feature quality by eliminating irrelevant and redundant features. The process of feature selection involves two primary components: a search mechanism responsible for exploring candidate feature subsets and an evaluation criterion that assesses the efficacy of these feature subsets.

The search space for possible subsets of a dataset with n original features is exponential, given by 2^n-1. As the number of features increases, the search space grows rapidly. Conducting an exhaustive search to explore all potential candidate subsets is theoretically desirable; however, it can be computationally intensive and time-consuming, often taking several days depending on the size of the search space [3]. To mitigate the computational burden, sequential forward and backward strategies (SFS and SBS [4]) provide an alternative approach. In SFS variants, features are incrementally added to an initially empty set until adding more features does not improve the criterion. Conversely, SBS variants begin with all features and successively eliminate each feature until no further improvement is achieved based on the criterion.

* Corresponding author at: Department of Software Engineering, Mehmet Akif Ersoy University, Burdur 15039, Turkey.
E-mail address: ehancer@mehmetakif.edu.tr (E. Hancer).

https://doi.org/10.1016/j.knosys.2023.111008
Received 13 June 2023; Received in revised form 31 August 2023; Accepted 13 September 2023
Available online 18 September 2023
0950-7051/© 2023 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Although sequential search strategies are more efficient than exhaustive search, they tend to converge in local optima. In contrast, evolutionary computation (EC) techniques offer a global search capability, making them better suited for exploring large search spaces compared to exhaustive and sequential search approaches. Among various EC techniques, particle swarm optimization (PSO) and genetic algorithms (GA) have been widely utilized for feature selection [3]. However, other EC techniques such as differential evolution (DE) potentially exhibit superior exploration abilities compared to PSO and GA in some tasks, e.g. numerical problems [5,6].

The complex interactions between features make feature selection challenging in addition to the large search space. Selecting only individually the most relevant features may result in redundancy because they may provide same information about the class label. There is a possibility that a subset of weakly related features can provide highly discriminatory information, even when each feature only provides limited information [7]. Feature interactions must be considered when evaluating a feature subset in order to reach an optimal result.

There are three main categories of feature selection depending on the evaluation criterion [8]: filter, wrapper and embedded. A filter method evaluates feature subsets independently of any classification algorithm, based on intrinsic data characteristics. In contrast, wrapper approaches use a classifier to evaluate the quality of the feature subset. Embedded approaches combine the selection of features and the training of a classifier into one learning process, so the selected features can be obtained from a classifier that has been trained. In spite of the fact that wrapper and embedded approaches perform better than filter methods in terms of the classification performance, filter approaches are the most efficient and more general to different classifiers, which is very important for large search spaces. Therefore, we focus on developing filter approaches in this paper.

Goals and contributions. There has been a growing effort in recent years to develop evolutionary approaches for feature selection. However, the majority of these approaches rely on wrapper evaluation, with only a small percentage focusing on evolutionary filter approaches. Despite some previous attempts to develop evolutionary filter approaches, a generalized feature selection approach is lacking in the literature [9]. For example, some evolutionary filter approaches based on standard information-theoretic [9] and rough set theory [10] are limited to working with binary-valued features, necessitating a discretization process for continuous-valued features. Moreover, some approaches may not adequately assess the dependency within a subset of features.

Neighborhood component feature selection (NCFS) [11,12] is a method that can be employed to select informative variables, regardless of the type of input data. By addressing an unconstrained multi-objective optimization problem, NCFS minimizes the mean loss of a neighborhood component analysis (NCA) regression model while simultaneously preventing overfitting through the utilization of a regularization term. The optimization process produces optimal weights, which are then used to determine the significance of the input variables. NCFS has been successfully applied in various domains, such as identifying relevant features in data captured with an RGB-D camera [13], reducing the dimensionality of input data to improve soil moisture forecasts [14], eliminating irrelevant features for spectroscopic analysis [15], and reducing the dimensionality of feature-space in microscopic images [16].

The objective function of NCFS incorporates a parameter λ, which represents the ratio of the NCA regression model's mean loss to the regularization term. Determining the λ involves computationally expensive processes like k-fold cross-validation and the grid search algorithm. Furthermore, the robustness of NCFS is compromised due to the varying optimal value of λ based on its boundary [17]. The last but not less important challenge is that the performance of NCFS highly depends on a user-specified threshold. To be specific, the final feature subset is determined by checking whether the optimal weight of a feature is higher than a user-specified threshold.

The overall goal of this study is to prevent the weaknesses of the NCFS method by introducing an evolutionary feature selection approach. To accomplish this, we introduce a filter evaluation criterion based on the optimal feature weights of NCFS and the number of features. Then, we develop an evolutionary approach for both single- and multi-objective scenarios based on this evolution criterion. It is worth noting that incorporating feature selection within a multi-objective framework usually yields superior classification performance compared to a single-objective approach since it explores the trade-offs between conflicting objectives without the requirement of a control parameter. This means that the algorithm can identify feature subsets that excel in one performance metric while not sacrificing too much in another. Also, the superiority of multi-objective scenario to single-objective scenario was also verified in several studies in the literature [10,18]. However, in the case of a multi-objective scenario, the user may need to select a single solution from a set of Pareto front solutions. Conversely, a single-objective scenario allows for the generation of a single solution. In this study, we specifically focus on both single-objective and multi-objective scenarios to provide researchers with flexibility in their selection process and to facilitate comparisons with recent studies. For the evolutionary search, among the various evolutionary algorithms available, we specifically choose the differential evolution (DE) algorithm [19] for the following reasons: (1) it requires only a few user-specified parameters, (2) it is simple to implement, (3) it has demonstrated successful search capabilities, and (4) it has been widely applied in feature selection with numerous successful applications [9,20–24]. The contributions of the paper can be summarized as follows:

1. We develop an evolutionary filter approach that allows the evolutionary process to prioritize features with high discriminative power, as determined by their optimal feature weights obtained by NCFS, while also considering the size of the feature subset. By incorporating these two factors, our approach ensures the selection of highly informative features while maintaining a reasonable subset size.
2. The introduced evolutionary approach serves as a viable alternative to the NCFS method, delivering excellent classification performance without requiring a user-specified parameter for determining the final feature subset.
3. The evolutionary approach based on the neighborhood-based filter criterion demonstrates its versatility by effectively handling both discrete and continuous-valued features in both single-objective and multi-objective scenarios. Its robustness and flexibility make it applicable across various contexts.
4. Our study marks the first integration of the nearest neighborhood strategy into an evolutionary feature selection approach, showcasing its novelty and innovative nature.

The rest of the paper is structured as follows. Section 2 provides a background on the DE algorithm, reviews recent related works, and explains the general structure of multi-objective optimization. Section 3 introduces the proposed evolutionary approach with two scenarios. Section 4 provides an outline of the experiment design. Section 5 discusses the experiment results by comparing the proposed filter approaches with a variety of approaches in the literature. Finally, Section 6 summarizes the study and discusses some open issues and future trends in feature selection.

2. Background

2.1. Differential evolution

Differential Evolution (DE) [19] is a stochastic optimization algorithm that can be used to find the global minimum or maximum of a task. It is a population-based algorithm that works by iteratively perturbing and improving a set of candidate solutions, referred to as

"individuals", until an optimal solution is found. DE does not require the objective function to be differentiable and can handle both continuous and discrete variables. It is particularly useful for complex, non-linear problems and is often used in engineering and scientific applications. The main steps of the DE algorithm are presented as follows:

1. Initialization: A set of candidate individuals, represented by vectors $\boldsymbol{x}_1, \ldots \boldsymbol{x}_i, \ldots, \boldsymbol{x}_{SN}$, are randomly generated by Eq. (1). Then, the fitness of each individual is evaluated using the objective function to be optimized, denoted as $f(x)$.

$$x_{ij} = x_j^{lb} + U_j(0,1) \times (x_j^{ub} - x_j^{lb}) \tag{1}$$

where x_{ij} is the jth variable of $\boldsymbol{x}_i = \{x_{i1}, \ldots, x_{ij}, \ldots, x_{iD}\}$ with D variables, x_j^{ub} and x_j^{lb} are the solution space boundaries for x_{ij}, and $U_j(0,1)$ is a uniformly generated number between 0 and 1.

2. Mutation: Three distinct individuals, denoted as $\boldsymbol{x}_{r1}$, $\boldsymbol{x}_{r2}$, and $\boldsymbol{x}_{r3}$, are chosen at random from the population. Then, a mutant solution, denoted as $\hat{\boldsymbol{x}}_i$, is generated by adding the weighted difference between two randomly selected parent individuals $\boldsymbol{x}_{r2}$ and $\boldsymbol{x}_{r3}$ to a third randomly selected parent individual $\boldsymbol{x}_{r1}$:

$$\hat{\boldsymbol{x}}_i = \boldsymbol{x}_{r1} + F \times (\boldsymbol{x}_{r2} - \boldsymbol{x}_{r3}) \tag{2}$$

where F is a weighting factor, controlling the amplitude of the difference between the parent individuals. A larger weighting factor will result in larger perturbations, increasing the chances of finding a global optimum, but also increase the risk of getting stuck in a local optimum.

3. Crossover: A new trial vector $\boldsymbol{u}_i$ is then generated by applying a crossover operation, such as the binomial crossover, between the mutant individual $\hat{\boldsymbol{x}}_i$ and the target individual $\boldsymbol{x}_i$, with a probability CR:

$$u_{ij} = \begin{cases} x'_{i,j}, U_j(0,1) < CR \\ x_{ij}, otherwise \end{cases} \tag{3}$$

where CR determines the likelihood that the mutant individual will be used to replace the target individual. A high crossover probability will lead to a more aggressive search, but also increases the risk of getting stuck in a local optimum.

4. Selection: The trial vector $\boldsymbol{u}_i$ is then compared to the target individual $\boldsymbol{x}_i$. If the trial vector has a better fitness value (as determined by the objective function), it replaces the target individual in the population.

The steps 2–4 are repeated until a stopping criterion is met. The final result of the algorithm is the individual with the best fitness value in the final population.

2.2. Multi-objective optimization

Multi-objective optimization (MOO) is a method of solving optimization problems with more than one conflicting objective that must be optimized simultaneously. MOO aims to find a set of Pareto optimal solutions that represent trade-offs between the different objectives. The characteristics of MOO include:

(1) Pareto Optimality: A solution is considered Pareto optimal if there is no other solution that can improve one objective without degrading at least one of the other objectives. These solutions form the non-dominated Pareto front.

(2) Diversity Preservation: To ensure the Pareto front is representative of the trade-offs, diversity preservation techniques such as crowding distance, niching, or diversity metrics are used.

(3) Robustness: Solutions obtained in MOO are often sensitive to changes in the problem, so robustness is important to maintain optimality under small changes. Robustness can be improved through sensitivity analysis.

2.3. Related works

Due to the motivation of our study, we focus on filter feature selection approaches. Filter approaches are a class of feature selection that evaluate feature subsets based on data intrinsic characteristics. These approaches are known for their efficiency, as they assess feature relevance independently of any learning model, making them computationally less expensive and easily applicable to large datasets.

One prominent approach is the Relief-based algorithms, such as ReliefF and RReliefF [25], which utilize nearest neighbors to estimate feature importance and handle both discrete and continuous data effectively. Laplacian Score [26] is a feature selection approach that measures the importance of features based on the graph Laplacian matrix and the distances between data points in a graph representation. Furthermore, the Chi-squared (χ^2) test [27] and Analysis of Variance (ANOVA) are statistical feature selection measures, which assess the independence between features and the target variable, making them suitable for handling categorical and continuous data, respectively. Consistency subset evaluation (CSE) [28] is a feature selection approach that assesses the value of a subset of attributes based on the consistency of class values when the training instances are projected onto that subset. The consistency of any subset is always at least as high as that of the full set of attributes.

One of the most popular categories of filter feature selection is mutual information (MI)-based approaches. Minimum Redundancy Maximum Relevance (mRmR)-based variants [29–31] designed to optimize the trade-off between feature relevance and redundancy. mRmR methods select features that are highly relevant to the target variable while minimizing the correlation among the selected features. Furthermore, the Information Gain (IG) criterion [32] quantifies the relevance of features by measuring the reduction in uncertainty they bring to the classification task. These approaches are simple to implement and computationally efficient but may not provide high performance in high-dimensional problems. Moreover, the user is required to specify the number of features, which can lead to a complex and computationally intensive process, especially with a large number of features in the dataset. To address this challenge, evolutionary computation (EC) techniques come into consideration in the last decades. EC techniques explore the feature subset space and find a subset of features that maximizes the MI between the features and the target variable. EC-based MI approaches are more computationally intensive than conventional MI approaches [23,24], but they are able to handle non-linear relationships between features and the target variable, as well as high-dimensional feature spaces. A variety of successful studies based on EC techniques [20,25,26] have recently been proposed by researchers in the literature. In terms of the performance, EC-based MI approaches tend to give better results than conventional MI approaches [27]. Despite its prominent success and high popularity, there are some drawbacks concerning the MI criterion [28]: (1) the definition of relevancy: MI-based feature selection approaches is their inability to handle cases where two features are individually informative but become redundant features together, leading to suboptimal feature subsets with excluded potentially valuable information; and (2) difficulty in handling categorical variables: MI-based feature selection approaches are typically used for discrete variables and may have difficulty in handling continuous variables.

NCFS [12] can be considered as an alternative to MI approaches due to a variety of successful applications in a variety of fields. Despite its success in a variety of applications, its performance highly depends on the following cases: (1) the λ regularization parameter representing the ratio of the NCA regression model's mean loss to the regularization needs to be optimally specified; however, it is a computationally intensive process; and (2) a user-specified threshold is required to determine the feature subset: however, selecting a specified number of top-ranked with high weights does not guarantee the optimal feature subset. The first case was considered by Kim et al. [17] by normalizing scales between mean loss and regularization terms. However, the second case is still an open issue.

3. Proposed approach

In this section, we first introduce the concept of NCFS, and then we thoroughly explain the proposed approach with single- and multi-objective scenarios.

3.1. Neighborhood Component Feature Selection (NCFS)

The NCFS method [12] utilizes the Neighborhood Component Analysis (NCA) [29] regression model to assess the significance of input variables. By obtaining optimal weights that minimize the mean loss of the NCA regression model, this study assigns these weights as indicators of input variable significance.

Let $S = \{(x_1, y_1), \ldots, (x_i, y_i), \ldots, (x_N, y_N)\}$ be a set of samples, where N is the total number of samples, x_i is a D-dimensional vector and y_i is the corresponding class label. The objective is to discover a weighting vector, denoted as $\boldsymbol{w}$, that facilitates the selection of a feature subset optimized for nearest-neighbor classification. In terms of feature weighting, the weighted distance between x_i and x_j is calculated by:

$$D_w(x_i, x_j) = \sum_{k=1}^{D} w_k^2 |x_{ik} - x_{jk}| \tag{4}$$

where w_k denotes the feature weight of the kth feature, and x_{ik} is the kth feature value of the ith sample.

To achieve successful nearest neighbor classification, it is often desirable to maximize the leave-one-out classification performance on the training set, denoted as T. However, optimizing this performance directly is challenging due to its non-differentiable nature when selecting the nearest neighbor as the classification reference point. As a practical approximation, an effective strategy involves determining the reference point using a probability distribution. The probability of x_i for selecting x_j as a reference point is calculated as follows:

$$p_{ij} = \frac{\kappa(D_w(x_i, x_j))}{\sum_{j=1, j\neq i}^{n} \kappa(D_w(x_i, x_j))} \tag{5}$$

where $\kappa(z) = exp(-z/\sigma)$ is a kernel function with an input parameter σ. The σ parameter has a crucial role on the probability of each point to be selected as the reference point such that if $\sigma = 0$, only the closest neighbor can be used as a reference point; on the other hand, if $\sigma = +\infty$, every point except the query point has the same chance of being selected. By using the above definition, we can determine the probability that the query point x_i will be correctly classified as follows:

$$p_i = \sum_{j,j} y_{ij} pij \tag{6}$$

where $y_{ij} = 1$ if $y_i = y_j$; $y_{ij} = 0$, otherwise. As a result, we can calculate the average leave-one-out probability of correct classification:

$$F(\boldsymbol{w}) = \frac{1}{N} \sum_i p_i = \frac{1}{N} \sum_i \sum_j y_{ij} p_{ij} \tag{7}$$

where $\boldsymbol{w} = \{w_1, \ldots, w_k, \ldots, w_D\}$ is a feature weighting vector. If $\sigma = 0$, $F(\boldsymbol{w})$ becomes the true leave-one-out classification accuracy. The regularized version of $F(\boldsymbol{w})$ is as follows:

$$F(\boldsymbol{w}) = \frac{1}{N} \sum_{i=1}^{N} \underbrace{\left[\sum_{j=1}^{N} y_{ij} p_{ij} - \lambda \sum_{k=1}^{D} w_k^2 \right]}_{F_i(\boldsymbol{w})} = \frac{1}{N} \sum_{i=1}^{N} F_i(\boldsymbol{w}) \tag{8}$$

where λ is the regularization parameter. If σ defined in $\kappa(z)$ is set to 1, finding the feature weight vector $\boldsymbol{w}$ for given λ can be expressed as the following optimization objective function:

$$\hat{\boldsymbol{w}} = \underset{w}{argmin}\ f(\boldsymbol{w}) = \underset{w}{argmin} \frac{1}{N} \sum_{i=1}^{N} f_i(\boldsymbol{w}) \tag{9}$$

where $f(\boldsymbol{w}) = -F(\boldsymbol{w})$ and $f_i(\boldsymbol{w}) = -F_i(\boldsymbol{w})$. Moreover, adding a constant to Eq. (9) does not change the argument of the minimum. Therefore, the objective function can be formulated by adding the constant 1, as follows:

$$\begin{aligned} \hat{\boldsymbol{w}} &= \underset{w}{argmin}\{1 + f(\boldsymbol{w})\} \\ &= \underset{w}{argmin} \left\{ \frac{1}{N} \sum_i^N \sum_{j=1}^N p_{ij} - \frac{1}{N} \sum_i^N \sum_{j=1}^N p_{ij} y_{ij} + \lambda \sum_{k=1}^D w_k^2 \right\} \\ &= \underset{w}{argmin} \left\{ \frac{1}{N} \sum_i^N \sum_{j=1}^N p_{ij}(1 - y_{ij}) + \lambda \sum_{k=1}^D w_k^2 \right\} \\ &= \underset{w}{argmin} \left\{ \frac{1}{N} \sum_i^N \sum_{j=1}^N p_{ij} I(y_i, y_j) + \lambda \sum_{k=1}^D w_k^2 \right\} \end{aligned} \tag{10}$$

where $I(y_i, y_j)$ is 1 if $y_i \neq y_j$; otherwise, $I(y_i, y_j)$ is 0. After obtaining the $\boldsymbol{w}$ vector, we apply min–max normalization on $\boldsymbol{w}$ as follows:

$$w_j = \frac{w_j - \min(\boldsymbol{w})}{\max(\boldsymbol{w}) - \min(\boldsymbol{w})} \tag{11}$$

3.2. Proposed neighborhood-based approach

We consider the proposed evolutionary approach from the perspective of both single- and multi-objective scenarios due to the following reasons: (1) The single-objective scenario simplifies the selection process of a feature subset by searching for the most optimal feature subset, but it may overlook the trade-offs or conflicts between different objectives; and (2) The multi-objective scenario may result in a more diverse set of features, which can lead to improved generalization performance and reduced overfitting; however, the user needs to determine the optimal subset from a number of nondominated solutions. Overall, the choice between single-objective and multi-objective feature selection depends on the complexity of the problem, and the preferences of the decision-makers involved in the feature selection process.

3.2.1. Approach for single-objective scenario (DENCA)

As described in Section 2, NCFS determines the significance of features based on the weights obtained from the optimization process. To determine the feature subset, a user-specified threshold is required. However, the interpretation and evaluation of feature importance can be subjective, and the relevance of features may vary depending on domain knowledge. Moreover, NCFS may struggle to handle redundant features effectively due to considering only features with high weights. This could potentially lead to suboptimal feature subsets. It is therefore necessary to find another way to deal with this problematic issue.

In this study, we design an objective function to determine the feature subset based on the feature weights of NCFS, defined as follows.

$$f(\boldsymbol{x_i}, \boldsymbol{w}) = \beta \sum_{x_{ij} > 0.5} w_j - (1 - \beta) \frac{size(\boldsymbol{s_i})}{size(\boldsymbol{x_i})} \tag{12}$$

where $\boldsymbol{x_i}$ is a candidate solution vector in the population, $\boldsymbol{w}$ represents the weight vector obtained by NCFS, $\boldsymbol{s_i}$ represents the feature subset of $\boldsymbol{x_i}$ where $x_{ij} > 0.5$, $size(\boldsymbol{s_i})$ and $size(\boldsymbol{x_i})$ respectively represent the number of the selected and all features, and β is a control parameter to keep the balance between the subset weight and the subset size.

Example. To clarify how the objective value is calculated, we provide the following example. Assume that a weight vector $\boldsymbol{w}$ obtained by NCFS for a dataset with 8 features is as follows:

$$\boldsymbol{w} = \{1.5e-05, 3.1e-06, 1.96, 0.3999, 1.0072, 0.7705, 0.8825, 0.9088\}$$

After applying min–max normalization on $\boldsymbol{w}$ by Eq. (11), the updated $\hat{\boldsymbol{w}}$ vector is defined as follows:

$$\hat{\boldsymbol{w}} = \{6.3e-06, 0, 1, 0.2039, 0.5137, 0.3930, 0.4501, 0.4635\}$$

Assume that the β parameter in Eq. (12) is set to 0.05, and a solution vector $\boldsymbol{x}_i$ in the population is defined as follows:

$$\boldsymbol{x}_i = \{0.8147, 0.9057, 0.1269, 0.9133, 0.6323, 0.0975, 0.2784, 0.5468\}$$

Then, the selected feature subset $S_i = \{1, 2, 4, 5, 8\}$, where x_{ij} is greater than 0.5; therefore, the sum of weights for S_i is calculated as 5.9297. Then, the cost of $f(\boldsymbol{x}_i, \boldsymbol{w})$ is $0.05 \times 5.9297 + 0.95 \times 5/8 = 0.3539$.

Algorithm 1: Pseudo code of DENCA

Input: SN, F, CR, MCN, run, β, $\boldsymbol{w}$
Output: $mean_{acc}$, std_{acc}

```
for r ← 1 to run do
    Initialization Stage
    for i ← 1 to SN do
        Initialize x_i by Eq. (1);
        Evaluate the fitness value of f(x_i, w) by Eq. (12);
    end
    Evolutionary Search Stage
    while iter < MCN do
        for i ← 1 to SN do
            Select x_r1, x_r2 and x_r3 from population;
            Apply mutation to obtain a mutant vector x̂_i by Eq. (2);
            Apply crossover to obtain a trial vector u_i by Eq; (3);
            Evaluate the fitness value of f(u_i, w) by Eq. (12);
            if f(u_i, w) > f(x_i, w) then
                u_i is kept in the population instead of x_i;
            end
        end
        iter ← iter + 1;
    end
    Subset Evaluation Stage
    Determine the best individual x_g in the population and set s_r = [ ];
    for j ← 1 to D do
        if x_gj > 0.5 then
            Add jth feature to s_r;
        end
    end
    Transform training and test sets based on s_r;
    Build a model on the reduced training set using a classifier;
    Verify the model on the reduced test set;
    Calculate the accuracy (acc^r) based on predictions;
end
Calculate mean accuracy (mean_acc) and standard deviation (std_acc);
```

The overall approach (DENCA) for the single-objective scenario is presented in Algorithm 1. In the initialization stage, initial solutions are randomly generated and their fitness values are calculated. Then, an evolutionary search is carried out by using DE operations to find an optimal solution. Finally, the best solution is chosen from the population according to the fitness evaluation. Training and test sets are then transformed based on the final feature subset, and a learning model is built on the transformed training set. Finally, the model is verified on the transformed test set in terms of classification accuracy. This approach is run for a specified number of independent runs since running an evolutionary algorithm multiple times can improve the reliability of the results by reducing the influence of random fluctuations and initial conditions. This can lead to more consistent and robust solutions. The final accuracy is determined by taking the mean value of corresponding acc^r scores, where $r = \{1, 2, \ldots, run\}$.

The computational complexity of the DENCA approach can be summarized as follows. The complexity of initialization stage is $O(SN \times D)$, where SN is the population size and D is the dimensionality of the dataset. The complexity of evolutionary search stage is $O(MCN \times SN \times D)$, where MCN is the maximum number of iterations. This is because all of the required operations, mutation, crossover and fitness evaluation have a complexity of $O(D)$. Finally, the complexity of subset evaluation stage depends on the number of features in the subset and the number of samples in the datasets, but it can be approximated as $O(SN \times D)$. Therefore, the overall complexity of the DENCA approach can be approximated as $O(MCN \times SN \times D + SN \times D + SN \times D)$, which simplifies to $O(MCN \times SN \times D)$.

3.2.2. Approach for multi-objective scenario (MODENCA)

DENCA uses two main objectives weighted in a single objective function through the β control parameter, which specifies the contribution degree of each objective. Nevertheless, the β parameter needs to be pre-determined and the optimal value is dataset-dependent. To get rid of the β parameter, we redesign Eq. (12) in the concept of multi-objective optimization:

$$F(\boldsymbol{x}_i, \boldsymbol{w}) = \left\{ \sum_{x_{ij}>0} w_j, size(\boldsymbol{s}_i) \right\} \tag{13}$$

The overall approach (MODENCA) for the multi-objective scenario is presented in Algorithm 2. We select the MODE algorithm [30] among a variety of multi-objective DE variants due to its simplicity and effectiveness. Furthermore, MODE evolves candidates in the same manner as the standard DE with the exception of the selection scheme which is based on dominance-based selection inspired by Lampinen's criterion [31]. For each independent run, candidate solutions in the population are evolved, and then their accuracy scores and subset sizes are collected into a predefined set. Once all independent runs are carried out, a non-dominated sorting scheme [32] is applied to this predefined set to obtain the Pareto front. The purpose of non-dominated sorting is to sort the solutions obtained from the evolutionary algorithm into different non-dominated fronts, where each front represents a set of solutions that are not dominated by any other solution in the population. The concept of domination is defined as follows: a solution A dominates another solution B if the values of the objectives for solution A are better than or equal to those of solution B for all objectives, and there is at least one objective where the value of A is strictly better than that of B. Lastly, the average front is obtained by taking the average of the accuracy for solutions with the same feature size in Q. The average front can be useful in analyzing the performance of a multi-objective optimization algorithm as it provides information about the consistency and stability of the results obtained from the algorithm.

DENCA and MODENCA have similar stages and complexities, but the additional computations required for multi-objective optimization make MODENCA more computationally intensive than DENCA. The difference in computational intensity is primarily due to the additional fitness function evaluations and objective value calculations needed in MODENCA to handle the multiple conflicting objectives.

4. Experiment design

4.1. Datasets

We choose 30 datasets for the experimental study, presented in Tables 1 and 2 in terms of the number of features, classes and samples. The size of the features in the datasets varies from 18 to 10 000. The datasets in Table 1 are used for a comparative analysis of various feature selection approaches, including information-theoretic, evolutionary, NCFS, and gradient-search filter methods. As information-theoretic approaches can handle only discrete variables, the datasets in Table 1 consist of discrete features exclusively. Furthermore, the datasets listed in Table 2 are used to compare our results with those reported in related studies [33,34]. These datasets predominantly include real-valued features and have small-sample sizes, enabling us to evaluate the performance of feature selection approaches from multiple perspectives.

Algorithm 2: Pseudo code of MODENCA

Input: SN, F, CR, MCN, run, β, $\boldsymbol{w}$
Output: PF,AF

```
Define an empty set Q = [ ];
for r ← 1 to run do
    Initialization Stage
    for i ← 1 to SN do
        Initialize x_i by Eq. (1);
        Evaluate the fitness value of f(x_i, w) by Eq. (13);
    end
    Evolutionary Search Stage
    while iter < MCN do
        for i ← 1 to SN do
            Select x_r1, x_r2 and x_r3 from population;
            Apply mutation to obtain a mutant vector x̂_i by Eq. (2);
            Apply crossover to obtain a trial vector u_i by Eq; (3);
            Evaluate the objective values of f(u_i, w) by Eq. (13);
            Apply dominance-based selection between u_i and x_i;
        end
        iter ← iter + 1;
    end
    Subset Evaluation Stage
    for i ← 1 to SN do
        Define s_i^r = [ ];
        for j ← 1 to D do
            if x_ij > 0.5 then
                Add jth feature to s_i^r;
            end
        end
        Transform training and test sets based on s_i^r;
        Build a model on the reduced training set using a predefined classifier;
        Verify the model performance on the reduced test set;
        Calculate the classification accuracy acc_i^r based on predictions;
        Add the pair of acc_i^r and size(s_i^r) to set Q;
    end
end
Apply non-dominated sorting on Q to get the Pareto front (PF);
Calculate the average front (AF) on Q;
```

Table 1
Datasets used in the first part of experiments.

Dataset	Features	Classes	Samples
Lymph	18	4	148
Spect	22	2	267
Leddisplay	24	10	1000
Dermatology	33	6	366
Soybean large	35	19	307
Chess	36	2	3196
Waveform	40	3	5000
Connect	42	3	3196
Promoter	57	2	106
Splice	60	3	3190
Optic	64	10	5620
Audiology	68	24	226
Coil2000	85	2	9000
DNA	180	2	3186
PCMAC	3289	2	1943
Relathe	4322	2	1427

4.2. Parameter settings

The parameter settings of the proposed approach are experimentally determined as follows. The population size is set to 30, and the characteristic parameters of DE, F and CR, are respectively set to 0.5 and 0.2 for both single- and multi-objective scenarios. The maximum number of evaluations for single- and multi-objective scenarios is set to 3000 and 5000, respectively. Moreover, the parameters of NCFS are determined as follows: the solver type for estimating feature weights is determined as the limited memory Broyden–Fletcher–Goldfarb–Shanno (LBFGS) algorithm; the FitMethod is defined as exact to perform fitting using all the data; the IterationLimit is respectively set to 50 and 250 for DENCA and MODENCA. In Eq. (12), the β parameter is set to 0.05 for datasets with more than 60 features and to 0.1 for datasets with less than 60 features. Furthermore, a total of 30 independent runs are conducted for each approach, i.e, the *run* parameter in Algorithms 1 and 2 is set to 30. The feature subset evaluation is conducted by k-nearest neighbors (KNN), where the k is set to 5, thereby denoting as 5NN.

Table 2
Datasets used in the second part of experiments.

Dataset	Features	Classes	Samples
WBCD	30	2	569
Ionosphere	34	2	351
Sonar	60	2	208
Hill	100	2	606
Colon	2000	2	62
Srbct	2308	4	83
Lung2	3312	5	203
Lymphoma	4026	3	62
Glioma	4433	4	50
MLL	5848	3	72
Prostate	6033	2	102
DLBCL	6285	2	77
Leukemia	7129	2	72
Arcene	10 000	2	200

In the comparisons conducted on the datasets listed in Table 1, 70% of the samples from each dataset are randomly chosen to serve as the training set and the remaining 30% as the test set. For the datasets in Table 2, we use 10% of the samples for testing and the remaining samples for training to make fair comparisons with the results reported in [33]. During the data division procedure, the distribution of samples across classes is also taken into account.

To conduct comparisons in the context of multi-objective scenario, we utilize the hypervolume indicator (HV) metric and the inverted generational distance (IGD) metrics. Since each multi-objective approach in our study adopts a different objective criterion, we unify the evaluation by using accuracy and the number of features as common criteria to calculate the HV and the IGD metrics. To perform the calculation of the HV metric, the accuracy and number of features obtained by each multi-objective approach are normalized to the range [0, 1]. In this setting, the ideal point is defined as (0, 0), the nadir point as (1.1, 1.1), and the reference point for HV is set to (1, 1) based on recommendations from [35,36]. To compute the IGD metric, we initially gather the non-dominated solutions achieved by all the compared multi-objective evolutionary approaches on the test dataset. These solutions are combined into a unified set, which is then normalized to the [0, 1] range. Following this, the standard non-dominated sorting technique is applied to the unified set to establish the actual "true" Pareto front.

4.3. Approaches used in experiments

We verify the performance of the proposed approach by comparing them with the following feature selection approaches:

(1) DEMI and MODEMI [24]: The DE-based single-objective and multi-objective approaches are based on information theory, inspired by the mutual information feature selection (MIFS) method [37]. The goal is to maximize the mutual relevancy between features and the target, while minimizing the mutual redundancy within the feature subset.

Table 3
Results of DENCA for different β parameter values.

		$\beta = 0.05$	$\beta = 0.075$	$\beta = 0.1$	$\beta = 0.25$	$\beta = 0.5$
Lymph	acc	73.81 ± 5.41	71.98 ± 3.04	**88.09 ± 3e–16**	85.56 ± 1.39	85.63 ± 0.43
	size	1	3.1	4.96	8.93	9.03
Spect	acc	**81.25 ± 0**	**81.25 ± 0**	**81.25 ± 0**	**81.25 ± 0**	**81.25 ± 0**
	size	1	1.07	1	1.07	1.03
Leddisplay	acc	95.98 ± 7.67	**100 ± 0**	**100 ± 0**	**100 ± 0**	**100 ± 0**
	size	4.77	5	5	6.97	7.1
Dermatology	acc	89.94 ± 3.61	96.22 ± 0.64	96.98 ± 0.51	**97.3 ± 0.44**	97.05 ± 0.46
	size	7.23	13.2	16.63	18.93	20.63
Soybean	acc	79.3 ± 3.03	86.14 ± 1.23	**86.45 ± 1.05**	83.25 ± 0.84	83.73 ± 1.01
	size	8.67	12.77	13.53	17.5	18
Chess	acc	93.33 ± 1.34	**95.73 ± 0.43**	95.48 ± 0.34	93.03 ± 1.14	92.79 ± 1.06
	size	17.67	22.83	23.23	27.3	27.6
Waveform	acc	81.61 ± 0.56	80.62 ± 0.78	**81.67 ± 0.77**	80.75 ± 0.77	80.88 0.66
	size	14.23	19.17	21.17	26.87	29.73
Connect	acc	**75.31 ± 0.5**	74.05 ± 0.29	74 ± 0.33	72.4 ± 0.49	72.15 ± 0.39
	size	22.2	26.43	29.53	35.63	36.07
Promoter	acc	86.44 ± 7.27	**87.56 ± 5.46**	87.33 ± 4.5	84.89 ± 4.53	86.56 ± 4.83
	size	9.63	10.53	11.57	12.37	12.77
Splice	acc	**78.84 ± 1.66**	74.39 ± 1.51	72.3 ± 1.89	68.46 ± 1.14	67.63 ± 1.28
	size	9.36	15.77	20.7	34.8	37.6
Optic	acc	**98.76 ± 0.23**	98.7 ± 0.14	98.68 ± 0.12	98.68 ± 0.09	98.69 ± 0.1
	size	36.5	41.03	42.53	46.7	48.87
Audiology	acc	**68.56 ± 3.72**	63.44 ± 2.88	62.05 ± 1.99	61.13 ± 2.32	61.13 ± 2.10
	size	9.5	11.27	12.67	14.57	14.57
Coil2000	acc	**93.63 ± 0.13**	93.6 ± 0.13	93.62 ± 0.09	93.6 ± 0.15	93.58 ± 0.13
	size	21.53	24.3	25.77	30.57	34.37
DNA	acc	**85.14 ± 1.1**	84.06 ± 1.12	83.49 ± 1.14	82.94 ± 1.26	82.37 ± 1.35
	size	69.9	77.37	84.2	94.4	104.7
PCMAC	acc	**75.84 ± 1.94**	75.58 ± 1.58	75.72 ± 1.47	75.76 ± 1.68	74.87 ± 1.77
	size	1553.1	1574.6	1581.83	1628.1	1655.7
Relathe	acc	80.35 ± 1.47	80.54 ± 1.37	80.05 ± 1.61	**80.80 ± 1.45**	80.42 ± 1.31
	size	2097.5	2126.8	2123.17	2171.1	2202.4

(2) DEMIRF and MODEMIRF [24]: The DE-based single-objective and multi-objective approaches are based on information theory, ReliefF [38] and Laplacian score [39]. The goal is to maximize mutual relevancy between features and the target, while eliminating low-ranked features through ReliefF and Laplacian score.

(3) MRMR [40]: This is the most widely applied information theoretic approach. The details of MRMR can be found in Section 2.3.

(4) Quasi-Newton (QN-method) [41]: This is a gradient-based method which utilizes information about the function's slope to guide the search towards the minimum.

(5) PSO-CFS, EA-CFS and HS-CSE [42]: While HS-CSE is a evolutionary consistency-based filter approach with the search strategy as harmony search algorithm, PSO-CFS and EA-CFS are correlation-based feature selection approaches with the search strategies as particle swarm optimization and evolutionary algorithm. These algorithms are implemented on the Weka platform.

(6) MODECFS: At the core of this multi-objective DE-based approach is Pearson correlation. This method seeks to maximize the correlation between features and the target, while concurrently minimizing the correlation within the feature subset.

(7) MODEKRL: The foundation of this multi-objective DE-based approach lies in kernel mutual information. It aims to maximize the mutual relevance between features and the target through a kernel-based criterion [43,44] while simultaneously minimizing the mutual relevancy within the feature subset.

(8) Other approaches: We also compare the proposed approach with a variety of state-of-the-art filter approaches, such as DEFRS [34], OFS [33], Fisher [45], ReliefF [38] and Laplacian [39].

5. Experiment results

5.1. Results of single-objective scenario

(a) The β control parameter analysis. Our in-depth analysis of the β control parameter in Table 3, which governs the balance between the subset weight and size in the feature selection process (Eq. (12)), has yielded valuable insights. As we vary the β value, we observe a consistent trend of increasing subset size. Specifically, for the Spect and Leddisplay datasets, the accuracy remains steady at 81.25% and 100%, respectively, regardless of the chosen β value. However, for datasets like Lymph, Dermatology, Soybean, Connect, and Promoter, the highest accuracy is achieved when $\beta = 0.1$. In contrast, the Optic, Coil2000, and PCMAC datasets maintain relatively stable accuracy levels across different β values. Moreover, for the Audiology and DNA datasets, the best accuracy is attained when $\beta = 0.05$. Based on these findings, we can determine appropriate β values for specific scenarios. For datasets with more than 60 features, $\beta = 0.05$ since DENCA generally demonstrates the best performance in this value. For the remaining datasets, we choose $\beta = 0.1$ since DENCA generally performs similarly but produces smaller feature subsets compared to other values.

(b) Classification performance analysis. The results of the DENCA approach are presented with a variety of filter-based feature selection approaches in Table 4 over 5NN in terms of the mean classification accuracy and the feature subset size. In Table 4, the best values are denoted in bold. Moreover, the symbols in the columns denoted as 'T' represent whether DENCA performs significantly better than the corresponding approach in terms of the Wilcoxon Rank Sum Test. While '+' ('−') indicates the superiority (inferiority) of DENCA over

Table 4
Comparisons for discrete datasets in single-objective scenario.

Dataset		DEMI	T	DEMIRF	T	MRMR	T	Newton	T	NCFS	T	PSO-CFS	T	EA-CFS	T	HS-CIE	T	DENCA
Lymph	acc	**88.09 ± 3e–16**	=	**88.09 ± 3e–16**	=	80.95	+	84.52 ± 6.4	+	85.71	+	**88.09**	=	**88.09**	=	83.33	+	**88.09 ± 3e–16**
	size	11		12		5		9		9		9		7		11		4.96
Spect	acc	78.75 ± 2e–16	+	80.00 ± 3e–16	+	81.25	=	80.75 ± 0.96	+	81.25	=	81.25	=	81.25	=	**82.5**	–	81.25
	size	14		14		4		11		1		12		8		18		1
Leddisplay	acc	88.88 ± 0.42	+	**100**	=	54.67	=	51.56 ± 16.06	=	**100**	=	**100**	=	**100**	=	82.33	+	**100**
	size	16		7		5		12		7		6		6		10		5
Dermatology	acc	97.49 ± 0.64	–	96.48 ± 0.87	=	97.14	=	90.89 ± 3.45	+	91.43	+	**98.1**	–	96.19	+	93.33	+	96.98 ± 0.56
	size	16		22.30		33		17		13		16		15		23		16.63
Soybean	acc	84.29 ± 1.98	+	85.92 ± 1.04	=	82.89	+	73.77 ± 5.93	+	82.89	+	85.53	+	81.58	+	80.26	+	**86.14 ± 1.42**
	size	22.4		21.23		30		18		17		11		8		15		13.16
Chess	acc	91.89 ± 0.61	+	87.64 ± 1.72	+	52.14	+	76.55 ± 10.02	+	76.54	+	76.94	+	76.33	+	90.72	+	**95.46 ± 0.47**
	size	15.46		21.23		3		10		18		5		6		15		13.16
Waveform	acc	80.26 ± 0.51	+	81.43 ± 0.64	+	79.73	+	75.33 ± 7.71	+	79.2	+	82	+	78.67	+	77.8	+	**95.46 ± 0.47**
	size	11.73		17.56		37		20		20		15		13		22		13.16
Connect	acc	73.66 ± 0.17	+	73.98 ± 0.31	=	70.40	+	71.31 ± 0.87	+	72.03	+	70.73	+	73.3	+	73.94	=	**74 ± 0.33**
	size	19.53		14.8		4		21		35		6		15		26		29.53
Promoter	acc	86.11 ± 4.8	+	87.44 ± 4.34	=	50	+	76.8 ± 7.68	+	**93.33**	–	90	–	66.67	+	83.33	+	88.22 ± 4.77
	size	16.16		14.6		11		29		10		6		4		17		11.3
Splice	acc	74.85 ± 1.85	+	75.59 ± 2.21	+	51.62	+	57.86 ± 5.89	+	67.08	+	69.59	+	72	+	54.75	+	**78.84 ± 1.66**
	size	14.76		14.06		46		30		36		28		19		14		9.36
Optic	acc	97.37 ± 0.40	+	97.63 ± 0.65	+	98.40	=	95.04 ± 1.51	+	98.52	+	98.64	+	96.14	+	88.9	+	**98.76 ± 0.23**
	size	25.6		22.96		62		32		43		36		27		16		36.5
Audiology	acc	64.2 ± 2.24	+	65.17 ± 2.69	+	58.46	+	52.56 ± 8.83	+	63.08	+	**75.38**	–	63.08	+	55.38	+	67.9 ± 3.23
	size	28.03		22.16		16		34		10		5		12		20		9.56
Coil2000	acc	**93.68 ± 0.12**	=	93.65 ± 0.11	=	93.85	–	93.83 ± 0.14	–	93.53	+	93.68	–	93.58	=	93.83	–	93.61 ± 0.13
	size	43.13		23.56		42		43		20		10		11		39		22.66
DNA	acc	82.90 ± 1.34	+	83.26 ± 1.72	+	63.07	+	72.36 ± 5.14	+	**87.02**	–	84.4	+	78.5	+	60.88	+	85.14 ± 1.1
	size	73.06		65.73		62		90		65		36		38		54		69.9
PCMAC	acc	72.77 ± 2.95	+	75.94 ± 1.62	=	72.16	+	71.86 ± 2.55	+	75.77	=	**79.21**	–	75.77	=	78.18	–	75.6 ± 1.72
	size	1529.33		1619.6		1698		1645		96		771		893		1675		1543.8
Relathe	acc	76.95 ± 2.26	+	79.24 ± 1.84	+	70.56	+	77.06 ± 2.1	+	63.08	+	75.93	+	57.94	+	77.34	+	**80.09 ± 1.69**
	size	2118.8		2186		722		2161		99		1294		10		2492		2099.7

the corresponding approach, '=' indicates no such significant difference between DENCA and the corresponding approach.

According to Table 4, DENCA generally has shown significantly better classification performance than DEMI. Moreover, DENCA also gets smaller feature subsets than DEMI in most cases. The classification performance of DENCA is also significantly or slightly better than DEMIRF. In the cases where DENCA and DEMIRF achieve similar classification performance, DENCA generally finds smaller feature subsets than DEMIRF. For instance, while both DENCA and DEMIRF obtain the same accuracy in the Lymp and Leddisplay datasets, the feature subsets selected by DENCA are smaller than those of DEMIRF. When compared to other EC-based filter approaches, DENCA demonstrates comparable or significantly superior performance. Specifically, DENCA achieves similar or better classification results than PSO-CFS in most datasets, and even in cases where their performances are comparable, DENCA consistently selects smaller feature subsets than PSO-CFS. Similarly, while HS-CSE outperforms DENCA in a few datasets, DENCA remains superior to EA-CFS across all datasets. These results indicate that DENCA effectively balances the reduction of features while maintaining accurate predictive capabilities.

(c) CPU time analysis. Table 5 displays the average CPU time results, measured in seconds, for DE-based approaches. Note that the CPU time does not include the computations required for filter measures before the evolutionary process. It only accounts for the time taken during the actual evolutionary search. The computational efficiency of all DE-based filter approaches is evident, with even the most computationally expensive method completing the process in a reasonable time. Even in high-dimensional datasets like PCMAC and Relathe, feature subset selection can be accomplished in just 2 to 5 s.

Table 5
CPU time results of DE-based approaches.

	DEMIFS	DEMIRF	DENCA
Lymph	0.1234	**0.0839**	0.0969
Spect	0.1589	0.0646	**0.0542**
Leddisplay	0.2307	0.0688	**0.0594**
Dermatology	0.2818	0.0776	**0.0578**
Soybean	0.2677	0.0708	**0.0625**
Chess	0.2286	0.0703	**0.0557**
Waveform	0.1641	**0.0729**	**0.0729**
Connect	0.3406	0.1094	**0.1065**
Promoter	0.2786	0.0927	**0.0901**
Splice	0.2057	0.1057	**0.0771**
Optic	0.4219	0.1292	**0.0927**
Audiology	0.4589	0.1083	**0.0823**
Coil2000	0.6849	0.1156	**0.0849**
DNA	1.0552	0.1182	**0.1062**
PCMAC	154.039	**2.5031**	2.5474
Relathe	272.5052	5.5318	**4.4932**

From Table 5, it can also be observed that DENCA consistently demonstrates faster computational times compared to DEMIFS and DEMIRF. Specifically, for most datasets, DENCA's CPU time is lower, indicating its efficiency in feature selection. While DEMIFS and DEMIRF also offer relatively reasonable CPU times, DENCA outperforms them, particularly in datasets like Lymph, Spect, Leddisplay, Dermatology, and Soybean, where the differences in CPU times are notable. It is worth noting that in certain datasets with larger feature spaces, such as PCMAC and Relathe, the CPU times for all approaches are comparatively higher, but even in these cases, DENCA remains competitive and does not significantly lag behind the other methods. Overall, the

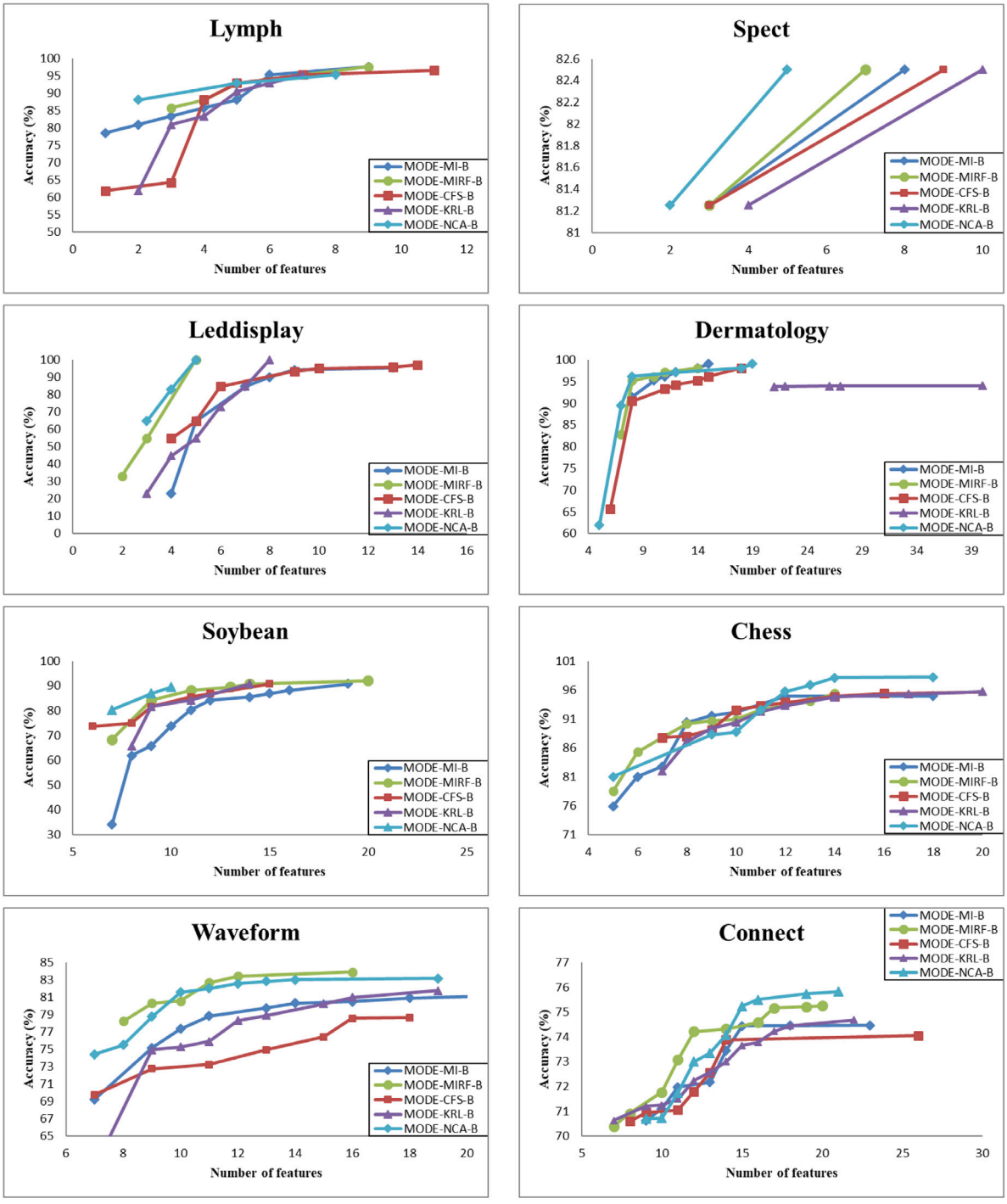


Fig. 1. Results of MODE-based approaches in terms of best Pareto front.

CPU time results highlight DENCA's ability to strike a balance between accurate feature selection and computational efficiency.

5.2. Results of multi-objective scenario

(a) Classification performance analysis. The results of MODENCA with multi-objective DE-based information-theoretic approaches are presented over 5NN in Figs. 1 and 2 in terms of the best (denoted as '-B') and the average (denoted as '-A') fronts, respectively. A detailed explanation of how we obtain the best and average fronts can be found in Algorithm 2. In terms of the best front, MODENCA provides similar or superior performance to the others in all datasets except for the Chess, Waveform and Promoter datasets. Particularly for datasets with more than 60 features, the performance gap between MODENCA and the other approaches is considerable: MODENCA obtains 78% accuracy with 930 features for the PCMAC dataset, while MODEMI and MODEMIRF obtain similar accuracy results with over 1430 features. The similar case can be also observed for the Relathe dataset. In terms of the average front, MODENCA generally performs better than the others except for the Audiology, Coil2000 and DNA datasets. Even in some datasets, the average front of MODENCA achieves better results than the best front of the others. It can therefore be stated that the

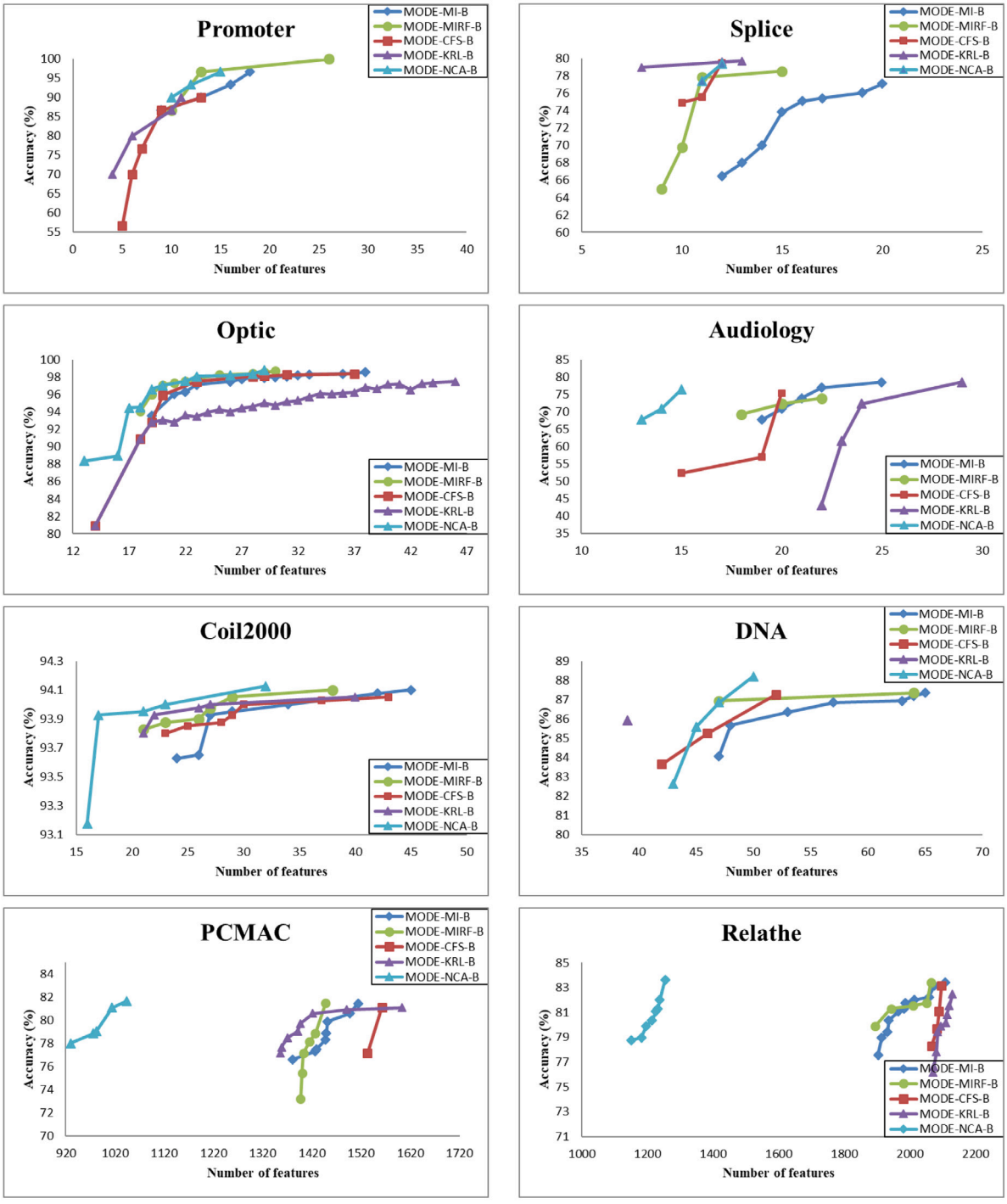


Fig. 1. (*continued*).

results of MODENCA are more consistent and stable than those of the other MODE-based approaches.

(b) HV metric analysis. The results presented in Table 6 demonstrate the HV metric values in terms of mean, standard deviation and significant test. The symbols denoted as 'T' in the columns have the same interpretation as in Table 4. A thorough examination of the HV values underscores the consistent superiority of MODENCA across a majority of datasets. This trend becomes even more pronounced when considering high-dimensional datasets like PCMAC and Relathe, where MODENCA's performance divergence from other methods is markedly significant. Notably, there are only three instances where MODENCA does not exhibit a statistically significant advantage. In summation, the proposed objective function not only excels within the context of single-objective scenario but also emerges as a promising and competitive alternative within the domain of multi-objective scenario.

(c) IGD metric analysis. The comparative performance analysis using the IGD metric is presented in Table 7, encompassing mean, standard deviation and significance testing across diverse datasets. The symbols labeled as 'T' in the columns retain the same interpretation as discussed in Table 6. The results highlight that MODENCA consistently equals or surpasses the performance of other approaches across datasets, with the exceptions of the Chess and Splice datasets. Particularly on high-dimensional datasets, MODENCA exhibits a substantial performance advantage over alternative algorithms. For instance, on the

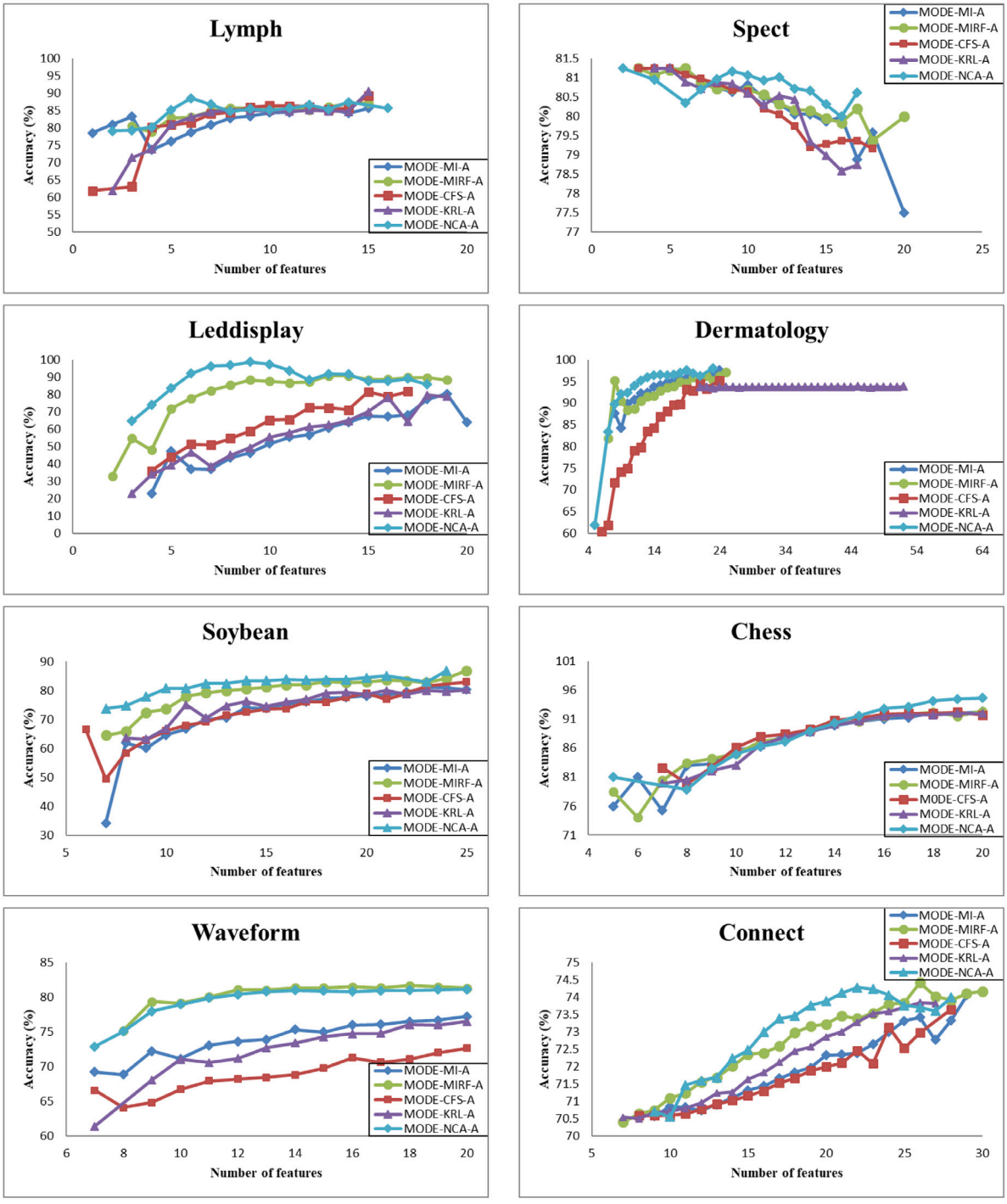


Fig. 2. Results of MODE-based approaches in terms of average Pareto front.

PCMAC dataset, MODENCA achieves a superior IGD of 0.1671, while other methods range between 0.24 and 0.29. Similarly, on the Relathe dataset, MODENCA achieves an IGD of 0.0317, whereas other methods score between 0.16 and 0.19. Among the others, MODEMIRF consistently demonstrates competitive results, while MODEKRL and MODEMI occasionally lag behind. These observations underscore MODENCA's consistent prominence, showcasing the lowest IGD values across various datasets and indicating its robust performance. These findings emphasize MODENCA's potential efficacy across a wide spectrum of datasets.

(d) CPU time analysis. Table 8 presents the mean CPU time results in seconds for MODE-based approaches. The findings suggest that incorporating feature selection in a multi-objective scenario demands more computational resources compared to a single-objective scenario. This increase in computational intensity can be attributed to the simultaneous optimization of conflicting objectives and the need for additional diversity maintenance operators. Despite the relatively higher time consumption compared to DE-based filter approaches, MODE-based filter approaches, particularly MODEMIRF and MODENCA, can complete the search process in less than half a minute. Among the MODE-based

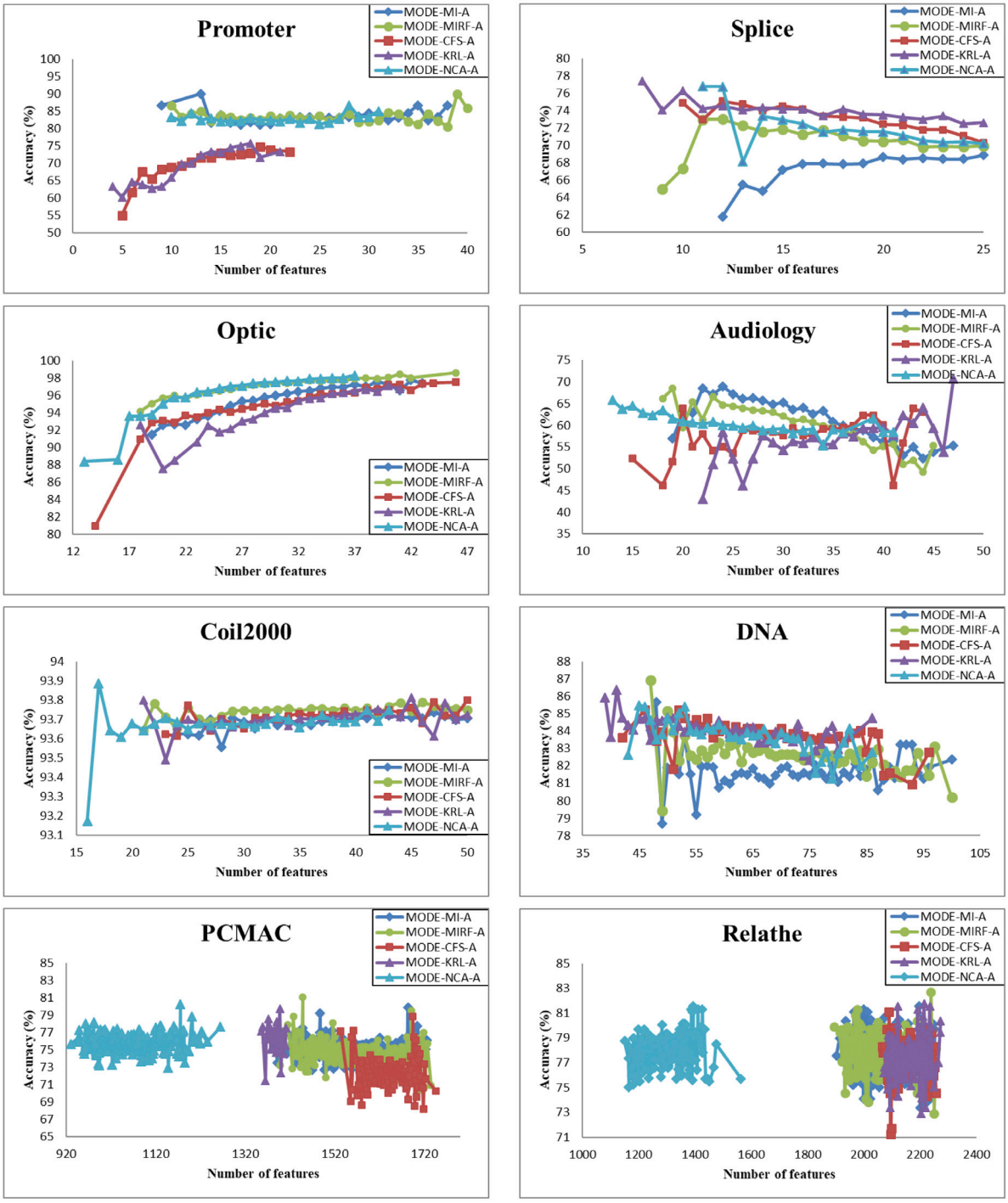


Fig. 2. (*continued*).

approaches, MODENCA consistently demonstrates lower CPU times, indicating its efficiency in performing the feature selection process. However, for the last five datasets, MODEMIRF outperforms MODENCA in terms of computational efficiency. It is essential to consider the trade-offs between computational complexity and the quality of feature subsets obtained. MODEKRL may not be the most efficient option for time-sensitive tasks. In conclusion, the proposed criterion shows promising efficiency for feature selection tasks, even in multi-objective optimization settings.

5.3. Further comparisons

Table 9 presents the results of DENCA and other state-of-the-art approaches on fourteen benchmark datasets. The evaluation is conducted using the 5NN classifier in terms of the average classification accuracy and the Wilcoxon rank-sum test. In the table, the best values are highlighted in bold. It is worth noting that, except for DENCA and DEFRS, all other approaches generate a unique feature subset, resulting in a single classification accuracy. Unfortunately, we are unable to

Table 6
Results of HV metric.

	MODEKRL	T	MODECFS	T	MODEMI	T	MODEMIRF	T	MODENCA
Lymph	0.6971 ± 0.0386	+	0.7300 ± 0.0310	+	0.6962 ± 0.0311	+	0.7066 ± 0.0439	+	**0.7510 ± 0.0398**
Spect	0.6358 ± 0.0361	+	0.6286 ± 0.0410	+	0.6396 ± 0.0419	+	0.6470 ± 0.0402	+	**0.6736 ± 0.0292**
Leddisplay	0.6461 ± 0.0450	+	0.6715 ± 0.0366	+	0.6282 ± 0.0393	+	0.7684 ± 0.0358	+	**0.7967 ± 0.0283**
Dermatology	0.6890 ± 0.0330	+	0.7143 ± 0.0263	+	0.6938 ± 0.0348	+	0.6950 ± 0.0346	+	**0.7397 ± 0.0218**
Soybean	0.6351 ± 0.0221	+	0.6568 ± 0.0293	+	0.6090 ± 0.0293	+	0.6651 ± 0.0288	+	**0.6860 ± 0.0264**
Chess	0.7327 ± 0.0205	−	0.7274 ± 0.0232	−	0.7356 ± 0.0274	−	**0.7591 ± 0.0246**	−	0.7109 ± 0.0313
Waveform	0.6050 ± 0.0242	+	0.6034 ± 0.0227	+	0.6195 ± 0.0232	+	0.6418 ± 0.0209	=	**0.6492 ± 0.0241**
Connect	0.5906 ± 0.0228	=	0.5849 ± 0.0211	=	0.5861 ± 0.0264	=	**0.6106 ± 0.0181**	−	0.5849 ± 0.0311
Promoter	0.6788 ± 0.0230	+	0.6242 ± 0.0374	+	0.6550 ± 0.0360	+	0.7060 ± 0.0229	+	**0.7349 ± 0.0260**
Splice	**0.6632 ± 0.0225**	−	0.6378 ± 0.0174	−	0.6511 ± 0.0149	−	0.6158 ± 0.0203	=	0.6154 ± 0.0208
Optic	0.6440 ± 0.0310	+	0.6745 ± 0.0286	+	0.6845 ± 0.0199	+	0.6760 ± 0.0264	+	**0.7225 ± 0.0229**
Audiology	0.4862 ± 0.0216	+	0.5025 ± 0.0288	+	0.5177 ± 0.0247	+	0.5479 ± 0.0286	+	**0.5594 ± 0.0259**
Coil2000	0.6931 ± 0.0211	+	0.6753 ± 0.0188	+	0.6564 ± 0.0208	+	0.7271 ± 0.0255	+	**0.7684 ± 0.0143**
DNA	0.6855 ± 0.0154	=	0.6427 ± 0.0173	+	0.6831 ± 0.0153	=	0.6722 ± 0.0160	+	**0.6866 ± 0.0104**
PCMAC	0.4980 ± 0.0067	+	0.4490 ± 0.0092	+	0.4986 ± 0.0059	+	0.4683 ± 0.1275	+	**0.6121 ± 0.0062**
Relathe	0.4634 ± 0.0052	+	0.4337 ± 0.1180	+	0.4879 ± 0.0054	+	0.4884 ± 0.0078	+	**0.6206 ± 0.0066**

Table 7
Results of IGD metric.

	MODEKRL	T	MODECFS	T	MODEMI	T	MODEMIRF	T	MODENCA
Lymph	0.1517 ± 0.0252	+	0.1322 ± 0.0210	+	0.1568 ± 0.0238	+	0.1523 ± 0.0183	+	**0.1235 ± 0.0231**
Spect	0.1067 ± 0.0388	+	0.1168 ± 0.0409	+	0.1049 ± 0.0375	+	0.0967 ± 0.0383	+	**0.0844 ± 0.0231**
Leddisplay	0.2457 ± 0.0467	+	0.1944 ± 0.0303	+	0.2531 ± 0.0372	+	0.1528 ± 0.0537	+	**0.1395 ± 0.0460**
Dermatology	0.1316 ± 0.0170	+	**0.0940 ± 0.0145**	−	0.1278 ± 0.0202	+	0.1224 ± 0.0153	+	0.1025 ± 0.0168
Soybean	0.1037 ± 0.0176	+	0.0978 ± 0.0189	+	0.1224 ± 0.0190	+	0.0675 ± 0.0144	+	**0.0513 ± 0.0112**
Chess	0.0590 ± 0.0106	−	0.0576 ± 0.0096	−	0.0565 ± 0.0088	−	**0.0486 ± 0.0070**	−	0.0711 ± 0.0130
Waveform	0.0883 ± 0.0151	+	0.1046 ± 0.0115	+	0.0743 ± 0.0158	+	0.0431 ± 0.0120	+	**0.0395 ± 0.0112**
Connect	0.0376 ± 0.0075	=	0.0412 ± 0.0093	+	0.0396 ± 0.0105	=	**0.0199 ± 0.0063**	−	0.0365 ± 0.0095
Promoter	**0.1023 ± 0.0143**	−	0.1262 ± 0.0213	+	0.1067 ± 0.0179	=	0.1637 ± 0.0176	+	0.1110 ± 0.0148
Splice	**0.0381 ± 0.0096**	−	0.0487 ± 0.0109	−	0.0431 ± 0.0105	−	0.0750 ± 0.0207	+	0.0667 ± 0.0180
Optic	0.0807 ± 0.0142	+	0.0582 ± 0.0122	+	0.0512 ± 0.0130	+	0.0493 ± 0.0080	+	**0.0337 ± 0.0053**
Audiology	0.2149 ± 0.0304	+	0.1684 ± 0.0300	+	0.1262 ± 0.0248	+	0.1015 ± 0.0213	+	**0.0871 ± 0.0196**
Coil2000	0.0612 ± 0.0182	+	0.0769 ± 0.0166	+	0.0947 ± 0.0208	+	0.0381 ± 0.0148	+	**0.0246 ± 0.0062**
DNA	0.0329 ± 0.0091	+	0.0692 ± 0.0158	+	0.0395 ± 0.0133	+	0.0353 ± 0.0102	+	**0.0284 ± 0.0074**
PCMAC	0.2466 ± 0.0064	+	0.2912 ± 0.0058	+	0.2463 ± 0.0049	+	0.2768 ± 0.1209	+	**0.1671 ± 0.0031**
Relathe	0.1895 ± 0.0032	+	0.1878 ± 0.0031	+	0.1627 ± 0.0059	+	0.1606 ± 0.0058	+	**0.0317 ± 0.0029**

Table 8
CPU time results of MODE-based approaches.

	MODEKRL	MODECFS	MODEMI	MODEMIRF	MODENCA
Lymph	38.5812	4.9714	8.4344	5.4193	**2.5609**
Spect	40.3328	5.0219	11.1646	8.1724	**5.3875**
Leddisplay	49.5443	**6.2313**	13.0604	9.825	6.6568
Dermatology	47.1604	7.9818	16.0057	11.5891	**7.4729**
Soybean	45.9781	10.1807	18.7036	13.4302	**9.6365**
Chess	46.0125	12.3271	20.8089	15.0484	**11.6896**
Waveform	50.4521	14.9427	22.9224	16.1297	**12.8318**
Connect	24.3927	23.2031	32.038	27.1318	**15.1309**
Promoter	51.7161	16.7630	23.3776	20.3458	**2.8281**
Splice	55.8307	18.9177	26.0469	21.624	**4.9943**
Optic	61.3693	22.2130	29.2854	23.9984	**6.374**
Audiology	64.8276	23.3385	29.8682	**5.8333**	7.9797
Coil2000	63.5224	24.6641	5.5677	**5.4307**	10.0948
DNA	64.4292	27.3656	8.4068	**5.7307**	12.2141
PCMAC	792.65	819.52	1510.8	**10.2083**	15.7422
Relathe	1178.63	1199.29	2930.17	**8.0589**	9.4062

provide information on the number of features obtained by each approach due to a lack of reporting in [33]. Therefore, a comparison through a multi-objective framework could not be performed. Furthermore, the symbols utilized in the columns marked with 'T' are identical to the ones employed in Tables 4 and 6.

According to the results presented in Table 9, it can be observed that DENCA consistently outperforms ReliefF, MI, Laplacian, LLC-FS, and FSV across the majority of datasets, with only one exception. When compared to INF, DNCA achieves significantly better classification performance in all datasets except for WBCD. The performance gap between DENCA and these approaches (MI, Laplacian, LLC-FS, and FSV) is exceptionally high. For instance, DENCA obtains nearly 81% classification accuracy in the Sonar dataset, while these approaches receive classification accuracies between 54% and 67%. Similar significant differences can be observed in other datasets as well. Therefore, the classification performance of these approaches cannot be considered adequate for continuous-valued datasets. Moreover, DENCA tends to outperform DEFRS in terms of average accuracy and achieves the highest accuracy in more datasets. However, DEFRS still performs well and achieves the highest accuracy on specific datasets. In summary, the proposed neighborhood-based approach achieves satisfactory classification performance not only in discrete-valued datasets but also in continuous-valued datasets.

6. Conclusions

In this study, we introduced an evolutionary filter approach to feature selection for both single- and multi-objective scenarios. To the best of our knowledge, this is the first study to integrate an NCA-based strategy into evolutionary feature selection. To be specific, we designed an objective function based on the feature weights of NCFS, and then integrated this objective function in the single-objective and multi-objective DE frameworks for feature selection. To assess the effectiveness of our proposed approach, we have conducted extensive comparisons with recently introduced evolutionary feature selection methods in the literature. The results indicate that our proposed approach outperforms these existing methods in terms of both single-objective and multi-objective scenarios. Furthermore, we have validated the effectiveness of our approach by comparing it with various state-of-the-art filter approaches. The outcomes demonstrate that the neighborhood search-based criterion can serve as a robust alternative to commonly used filter criteria, such as NCFS, ReliefF, Laplacian Score, and mutual information.

Table 9
Results of DENCA with state-of-the-art approaches.

Dataset	DENCA	DEFRS	OFS	Fisher	PCC	ReliefF	MI	Laplacian	INF	LLC-FS	FSV
Ionosphere	**91.2 ± 1.8**	88.79 ± 3.38	89.43	88	87.43	82.29	79.43	83.43	81.71	85.43	79.43
		+	+	+	+	+	+	+	+	+	+
WBCD	93.1 ± 0.64	93.74 ± 2.19	**93.85**	93.32	93.32	89.62	90.16	92.44	93.67	90.86	90.16
		=	=	=	=	+	+	+	=	+	+
Sonar	**81.67 ± 5.62**	73 ± 9.05	68.27	71.08	71.08	67.79	54.79	59.55	69.55	66.84	54.79
		+	+	+	= +	+	+	+	+	+	+
Hill	**69.89 ± 1.85**	59.5 ± 3	59.5	53.14	53.14	55.7	55.45	49.67	50.33	54.79	55.45
		+	+	+	+	+	+	+	+	+	+
Colon	**98.88 ± 4.22**	**98.88 ± 4.22**	75	73.33	73.33	71.67	51.67	51.67	56.67	61.67	51.67
		=	+	+	+	+	+	+	+	+	+
Srbct	**100 ± 0**	97.91 ± 4.73	88.33	88.33	58.33	86.67	63.33	31.67	26.67	38.33	63.33
		+	+	+	+	+	+	+	+	+	+
Lung2	**99.83 ± 0.91**	**99.83 ± 0.91**	93	79	81.5	80	74.5	76.5	81	79	83.5
		=	+	+	+	+	+	+	+	+	+
Lymphoma	**100 ± 0**	**100 ± 0**	**100**	98.33	95	96.67	78.33	93.33	60	95	78.33
		=	=	+	+	+	+	+	+	+	+
Glioma	**100 ± 0**	**83.33 ± 5.6e−16**	68	64	66	26	56	42	40	48	62
		+	+	+	+	+	+	+	+	+	+
MLL	**99.56 ± 1.69**	99.33 ± 2.03	92.86	90	78.57	92.86	64.29	84.29	91.43	87.14	82.86
		=	+	+	+	+	+	+	+	+	+
Prostate	81.33 ± 11.06	80.66 ± 7.39	**93**	91	91	91	63	59	48	72	63
		=	−	−	−	−	+	+	+	+	+
DLBCL	92.05 ± 5.53	90.76 ± 4.23	**95**	79.25	44.25	81.25	60.5	68.5	73.75	76.25	78.75
		+	−	+	+	+	+	+	+	+	+
Leukemia	**99.04 ± 3.62**	**99.04 ± 3.62**	95.71	87.14	87.14	84.29	60	58.57	47.14	47.14	60
		=	+	+	+	+	+	+	+	+	+
Arcene	**90.48 ± 0**	90.31 ± 0.86	86	65.5	65.5	67.5	58.5	70	71.5	65.5	69
		=	+	+	+	+	+	+	+	+	+
Avg. Accuracy	**92.64**	89.64	85.56	80.10	74.39	76.66	64.99	65.75	63.67	69.13	69.44

In our future work, we plan to extend our research by developing a filter criterion specifically tailored for feature selection in multi-label classification problems. Multi-label classification is known to be more challenging than conventional classification tasks, and thus, a dedicated filter criterion would be valuable in addressing this complexity.

CRediT authorship contribution statement

Emrah Hancer: Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization. **Bing Xue:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Mengjie Zhang:** Writing – review & editing, Writing – original draft, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

References

[1] E. Hancer, B. Xue, D. Karaboga, M. Zhang, A binary ABC algorithm based on advanced similarity scheme for feature selection, Appl. Soft Comput. 36 (2015) 334–348, http://dx.doi.org/10.1016/j.asoc.2015.07.023.

[2] P. Wang, B. Xue, J. Liang, M. Zhang, Feature selection using diversity-based multi-objective binary differential evolution, Inform. Sci. 626 (2023) 586–606, http://dx.doi.org/10.1016/j.ins.2022.12.117.

[3] B.H. Nguyen, B. Xue, M. Zhang, A survey on swarm intelligence approaches to feature selection in data mining, Swarm Evol. Comput. 54 (2020) 100663, http://dx.doi.org/10.1016/j.swevo.2020.100663.

[4] F. Ferri, P. Pudil, M. Hatef, J. Kittler, Comparative study of techniques for large-scale feature selection, in: E.S. Gelsema, L.S. Kanal (Eds.), Pattern Recognition in Practice IV, in: Machine Intelligence and Pattern Recognition, Vol. 16, North-Holland, 1994, pp. 403–413, http://dx.doi.org/10.1016/B978-0-444-81892-8.50040-7.

[5] M. Iwan, R. Akmeliawati, T. Faisal, H.M. Al-Assadi, Performance comparison of differential evolution and particle swarm optimization in constrained optimization, Procedia Eng. 41 (2012) 1323–1328, http://dx.doi.org/10.1016/j.proeng.2012.07.317, International Symposium on Robotics and Intelligent Sensors 2012 (IRIS2012).

[6] D. Zhang, B. Wei, Comparison between differential evolution and particle swarm optimization algorithms, in: IEEE International Conference on Mechatronics and Automation, 2014, pp. 239–244, http://dx.doi.org/10.1109/ICMA.2014.6885702.

[7] M. García-Torres, R. Ruiz, F. Divina, Evolutionary feature selection on high dimensional data using a search space reduction approach, Eng. Appl. Artif. Intell. 117 (2023) 105556, http://dx.doi.org/10.1016/j.engappai.2022.105556.

[8] P. Wang, B. Xue, J. Liang, M. Zhang, Feature selection using diversity-based multi-objective binary differential evolution, Inform. Sci. 626 (2023) 586–606, http://dx.doi.org/10.1016/j.ins.2022.12.117.

[9] E. Hancer, A multi-objective differential evolution feature selection approach with a combined filter criterion, in: 2nd International Symposium on Multidisciplinary Studies and Innovative Technologies (ISMSIT2018), 2018, pp. 1–8.

[10] B. Xue, L. Cervante, L. Shang, W.N. Browne, M. Zhang, A multi-objective particle swarm optimisation for filter-based feature selection in classification problems, Connect. Sci. 24 (2–3) (2012) 91–116, http://dx.doi.org/10.1080/09540091.2012.737765.

[11] J. Goldberger, S. Roweis, G. Hinton, R. Salakhutdinov, Neighbourhood components analysis, in: Proceedings of the 17th International Conference on Neural Information Processing Systems, NIPS '04, MIT Press, Cambridge, MA, USA, 2004, pp. 513–520.

[12] W. Yang, K. Wang, W. Zuo, Neighborhood component feature selection for high-dimensional data, J. Comput. 7 (1) (2012) http://dx.doi.org/10.4304/jcp.7.1.161-168.

[13] M. Paolanti, L. Romeo, D. Liciotti, R. Pietrini, A. Cenci, E. Frontoni, P. Zingaretti, Person re-identification with RGB-D camera in top-view configuration through multiple nearest neighbor classifiers and neighborhood component features selection, Sensors 18 (10) (2018) http://dx.doi.org/10.3390/s18103471.

[14] R. Prasad, R.C. Deo, Y. Li, T. Maraseni, Ensemble committee-based data intelligent approach for generating soil moisture forecasts with multivariate hydro-meteorological predictors, Soil Tillage Res. 181 (2018) 63–81, http://dx.doi.org/10.1016/j.still.2018.03.021.

[15] J. Zhang, Y. Xiong, S. Min, Bootstrapping search margin-based nearest neighbor method for qualitative spectroscopic analysis, Spectrochim. Acta A 223 (2019) 117110, http://dx.doi.org/10.1016/j.saa.2019.05.015.

[16] A. Nawandhar, N. Kumar, R. Veena, L. Yamujala, Stratified squamous epithelial biopsy image classifier using machine learning and neighborhood feature selection, Biomed. Signal Process. Control 55 (2020) 101671, http://dx.doi.org/10.1016/j.bspc.2019.101671.

[17] H. Kim, T.H. Lee, T. Kwon, Normalized neighborhood component feature selection and feasible-improved weight allocation for input variable selection, Knowl.-Based Syst. 218 (2021) 106855, http://dx.doi.org/10.1016/j.knosys.2021.106855.

[18] B. Xue, M. Zhang, W.N. Browne, Particle swarm optimization for feature selection in classification: A multi-objective approach, IEEE Trans. Cybern. 43 (6) (2013) 1656–1671, http://dx.doi.org/10.1109/TSMCB.2012.2227469.

[19] R. Storn, K. Price, Differential evolution – A simple and efficient heuristic for global optimization over continuous spaces, J. Global Optim. 11 (1997) 341–359, http://dx.doi.org/10.1023/A:1008202821328.

[20] E. Hancer, Differential evolution for feature selection: a fuzzy wrapper–filter approach, Soft Comput. 23 (13) (2019) 5233–5248, http://dx.doi.org/10.1007/s00500-018-3545-7.

[21] E. Hancer, A new multi-objective differential evolution approach for simultaneous clustering and feature selection, Eng. Appl. Artif. Intell. 87 (2020) 103307, http://dx.doi.org/10.1016/j.engappai.2019.103307.

[22] E. Hancer, B. Xue, M. Zhang, Fuzzy filter cost-sensitive feature selection with differential evolution, Knowl.-Based Syst. 241 (2022) 108259, http://dx.doi.org/10.1016/j.knosys.2022.108259.

[23] E. Hancer, A differential evolution approach for simultaneous clustering and feature selection, in: International Conference on Artificial Intelligence and Data Processing (IDAP2018), 2018, pp. 1–7.

[24] E. Hancer, B. Xue, M. Zhang, Differential evolution for filter feature selection based on information theory and feature ranking, Knowl.-Based Syst. 140 (2018) 103–119, http://dx.doi.org/10.1016/j.knosys.2017.10.028.

[25] E. Hancer, B. Xue, M. Zhang, D. Karaboga, B. Akay, Pareto front feature selection based on artificial bee colony optimization, Inform. Sci. 422 (2018) 462–479, http://dx.doi.org/10.1016/j.ins.2017.09.028.

[26] R. Espinosa, F. Jiménez, J. Palma, Surrogate-assisted and filter-based multiobjective evolutionary feature selection for deep learning, IEEE Trans. Neural Netw. Learn. Syst. (2023) 1–15, http://dx.doi.org/10.1109/TNNLS.2023.3234629.

[27] B. Tran, B. Xue, M. Zhang, A new representation in PSO for discretization-based feature selection, IEEE Trans. Cybern. 48 (6) (2018) 1733–1746, http://dx.doi.org/10.1109/TCYB.2017.2714145.

[28] J.R. Vergara, P.A. Estevez, A review of feature selection methods based on mutual information, Neural Comput. Appl. 24 (2014) 175–186, http://dx.doi.org/10.1007/s00521-013-1368-0.

[29] J. Goldberger, S. Roweis, G. Hinton, R. Salakhutdinov, Neighbourhood components analysis, in: Proceedings of the 17th International Conference on Neural Information Processing Systems, NIPS '04, MIT Press, Cambridge, MA, USA, 2004, pp. 513–520.

[30] K. Price, R.M. Storn, J.A. Lampinen, Differential Evolution: A Practical Approach To Global Optimization (Natural Computing Series), Springer-Verlag, Berlin, Heidelberg, 2005.

[31] J. Lampinen, Solving problems subject to multiple nonlinear constraints by differential evolution, in: 7th International Conference on Soft Computing, 2001, pp. 50–57.

[32] K. Deb, S. Agrawal, A. Pratap, T. Meyarivan, A fast elitist non-dominated sorting genetic algorithm for multi-objective optimization: NSGA-II, in: M. Schoenauer, K. Deb, G. Rudolph, X. Yao, E. Lutton, J.J. Merelo, H.-P. Schwefel (Eds.), Parallel Problem Solving from Nature PPSN VI, Springer Berlin Heidelberg, Berlin, Heidelberg, 2000, pp. 849–858.

[33] P. Zhou, X. Hu, P. Li, X. Wu, OFS-Density: A novel online streaming feature selection method, Pattern Recognit. 86 (2019) 48–61, http://dx.doi.org/10.1016/j.patcog.2018.08.009.

[34] E. Hancer, New filter approaches for feature selection using differential evolution and fuzzy rough set theory, Neural Comput. Appl. 32 (2020) 2929–2944, http://dx.doi.org/10.1007/s00521-020-04744-7.

[35] H. Xu, B. Xue, M. Zhang, A duplication analysis-based evolutionary algorithm for biobjective feature selection, IEEE Trans. Evol. Comput. 25 (2) (2021) 205–218, http://dx.doi.org/10.1109/TEVC.2020.3016049.

[36] R. Jiao, B. Xue, M. Zhang, Benefiting from single-objective feature selection to multiobjective feature selection: A multiform approach, IEEE Trans. Cybern. (2022) 1–14, http://dx.doi.org/10.1109/TCYB.2022.3218345.

[37] R. Battiti, Using mutual information for selecting features in supervised neural net learning, IEEE Trans. Neural Netw. 5 (4) (1994) 537–550, http://dx.doi.org/10.1109/72.298224.

[38] I. Kononenko, Estimating attributes: Analysis and extensions of RELIEF, in: F. Bergadano, L. De Raedt (Eds.), Machine Learning: ECML-94, Springer Berlin Heidelberg, Berlin, Heidelberg, 1994, pp. 171–182.

[39] X. He, D. Cai, P. Niyogi, Laplacian score for feature selection, in: Proceedings of the 18th International Conference on Neural Information Processing Systems (NIPS'05), MIT Press, 2005, pp. 507–514.

[40] H. Peng, F. Long, C. Ding, Feature selection based on mutual information criteria of max-dependency, max-relevance, and min-redundancy, IEEE Trans. Pattern Anal. Mach. Intell. 27 (8) (2005) 1226–1238.

[41] T. Coleman, Y. li, On the convergence of reflective Newton methods for large-scale nonlinear minimization subject to bounds, Math. Program. 67 (1994) 189–224, http://dx.doi.org/10.1007/BF01582221.

[42] I.H. Witten, E. Frank, M.A. Hall, C.J. Pal, Data Mining, Fourth Edition: Practical Machine Learning Tools and Techniques, fourth ed., Morgan Kaufmann Publishers Inc., San Francisco, CA, USA, 2016.

[43] Y.-I. Moon, B. Rajagopalan, U. Lall, Estimation of mutual information using kernel density estimators, Phys. Rev. E 52 (1995) 2318–2321, http://dx.doi.org/10.1103/PhysRevE.52.2318.

[44] E. Hancer, Fuzzy kernel feature selection with multi-objective differential evolution algorithm, Connect. Sci. 31 (4) (2019) 323–341, http://dx.doi.org/10.1080/09540091.2019.1639624.

[45] R.A. Fisher, The use of multiple measurements in taxonomic problems, Ann. Eugen. 7 (2) (1936) 179–188, http://dx.doi.org/10.1111/j.1469-1809.1936.tb02137.x, arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1469-1809.1936.tb02137.x.